

Project Synopsis: Analysis of Cost Overrun Factors in Construction Projects

1. Overview and Objective:

This project examines a dataset on the causes of cost overruns in construction projects. Its main aim is to identify, group, and measure the major factors behind cost overruns and to develop a risk assessment model based on these findings.

2. Data Source:

The study uses survey data from 143 construction industry professionals, including Project Managers, Engineers, clients, consultants, and contractors. Respondents evaluated the effect of 44 predefined factors on project cost overruns using a rating scale.

3. Key Analysis Performed:

Factor Analysis: The main analytical method was factor analysis, which reduced 44 separate risk variables into a smaller set of underlying latent factors.

Factor Extraction: The analysis identified 6 significant factors explaining more than 26% of the shared variance in the dataset. These factors were rotated using Varimax to improve interpretation.

Factor Interpretation: Based on high-loading variables, the six extracted factors were interpreted as: construction execution/site management problems, design and planning deficiencies, financial and administrative delays, scope/client changes, supply and procurement issues, and contract management/external conditions.

Regression Analysis: A multiple regression model was developed to predict cost overrun frequency. The model used the six extracted factors, as well as respondents' qualifications and experience, as predictor variables.

Key Findings from Regression:

Strongest Positive Predictors: Factor 5 (supply and procurement issues) and Factor 6 (contract management/external conditions) were the strongest contributors to higher cost overrun frequency.

Strongest Negative Predictor: Relevant working experience was the most significant negative predictor, showing that more experienced professionals reported fewer cost overruns. Factor 4 (scope/client changes) also showed a significant negative effect.

Other Notable Predictors: Highest formal qualification showed a marginally significant positive effect, while Factors 1, 2, and 3 were not statistically significant in the model.

Risk Score and Segmentation: A composite risk score was calculated for each respondent using regression coefficients, creating a single measure of overall risk exposure. Respondents were then divided into 5 risk groups. The analysis also compared the profiles of the top 3 highest-risk respondents and the bottom 3 lowest-risk respondents.

4. Summary of Conclusions:

The findings show that the main causes of cost overruns in the surveyed construction projects are

supply chain issues and contract management/external conditions. Experience emerged as the strongest protective factor, suggesting that practical knowledge plays a major role in reducing cost overrun risk. The six-factor framework gives a more structured understanding of project cost overruns, while the risk scoring and segmentation provide a practical way to identify higher-risk project profiles.

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